METHOD 19 / A COMPLETE BEGINNER GUIDE

Tail dependence

Question this method answers: When one FX return is in an extreme tail, how often is the other in its tail too?

Tail dependence is not one Pearson r. It is a conditional frequency in the joint tails of uniform transforms u and v built from returns.

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Made-up event counts only — no live pair tail estimates from snapshots.

Prepared for: 19 - Tail Dependence. README status: not started.

02 / THE BASIC IDEA

Extreme days together

Pearson r averages co-movement across all days. Tail dependence asks about *worst* days: if GBP/USD has a very low return, is EUR/USD also very low that day or soon after?

Condition on one tail

L■(α)	Share of conditioning-tail dates where both legs are in the chosen tail.
λL	Theoretical limit as α → 0 for lower-tail co-crash probability.
Not the same	A 5% cutoff estimate is finite-sample; it is not automatically λL.

03 / RETURNS BEFORE TAILS

FX percent math (majors)

```
Simple return = P_t / P_{t-1} - 1
Log return = ln(P_t / P_{t-1})
```

1.1000 → 1.1110 is about +1% simple return. Tail events are defined on return distributions — usually after converting each leg to ranks or fitted CDF values u and v in (0,1).

A –1% day is not automatically “in the 5% tail” unless your rank or model says so. Keep the return type and window fixed.

04 / WHERE A LIVES

Uniforms u and v

```
u_i = F_hat_X(x_i)
v_i = F_hat_Y(y_i)
```

F_hat can be empirical ranks, a GARCH-filtered residual CDF, or another margin model. The tail cutoff α (such as 0.05) applies to these uniforms, not to raw percent returns.

Section 19 in forex-formulas-explained.pdf matches this repo placement. Method 18 (copulas) uses the same u,v building blocks with a different summary object.

05 / PATTON (2006)

Research notes

Modelling asymmetric exchange rate dependence. International Economic Review, 47(2), 527–556. DOI: 10.1111/j.1468-2354.2006.00387.x.

Shows FX dependence is often asymmetric across tails — joint crashes can differ from joint rallies. Copula tail parameters and empirical tail frequencies both target that asymmetry; this method focuses on countable tail events.

Future experiment outputs should cite the same margin pipeline used to build u and v.

Symbol dictionary

Symbol	Meaning
u, v	Uniform transforms of X and Y returns on each date.
α (alpha)	Small tail threshold on [0,1], e.g. 0.05.
L■(α)	Empirical conditional tail frequency (see formula page).
λL	Lower-tail dependence coefficient; limit as α → 0.
λU	Upper-tail analogue using u ≥ 1–α, v ≥ 1–α.
Event counts	Numerator and denominator counts — always report them.

07 / COUNTING RULE

Finite-sample estimator L■

```
L■(α) = (# dates with u ≤ α AND v ≤ α)
/ (# dates with v ≤ α)
```

The conditioning series here is v (formula sheet convention). Swapping roles changes the ratio unless dependence is symmetric.

Each counted date is one row in your aligned return panel. Missing quotes drop that row before ranks are formed.

This frequency is a percent

When the denominator is 20 dates, the numerator 12 means 60% of those conditioning-tail dates also had u in the lower tail at the chosen α. That percentage applies to the tail subset — not to all calendar days.

Asymptotic λL

```
λL = lim α→0 P(u ≤ α | v ≤ α)
```

λL describes the extreme lower tail under a copula or probability model. With finite data, α = 0.05 gives a stable teaching cutoff but estimates a thicker tail region than the limit.

Do not report L■(0.05) = 0.60 as if it were λL without explanation. Show α, counts, and the margin method used for u and v.

09 / 12 OF 20 JOINT LOWER-TAIL DAYS

Teaching example

Quantity	Value	Plain reading
α cutoff	0.05 (illustrative)	Lower-tail threshold on uniforms
Dates with $v \le \alpha$	20	Conditioning-tail event count (denominator)
Also $u \le \alpha$	12	Joint lower-tail event count (numerator)
$L_{\blacksquare}(\alpha)$	12/20 = 0.60	Conditional frequency among v-tail dates

If 12 of 20 GBP/USD lower-tail dates also have EUR/USD in its lower tail, the illustrative conditional frequency is 0.60. That is 60% of the conditioning-tail dates — not 60% of all trading days in the window.

Repeat the count table for upper tails or opposite tails when your FX question needs them.

Report like a lab note

Minimum reporting block

1. Sample dates and pair names. 2. Return type (simple/log). 3. How u and v were built. 4. α and which leg conditions. 5. Numerator, denominator, and L■. 6. Whether the number is a teaching cutoff or an extrapolated λ estimate.

A table beats a lone decimal: readers can see whether the denominator is large enough.

Versus Pearson and copulas

Pearson r can be high while tail co-crash frequency is low (Gaussian-like margins). Copula θ may encode tail shape but still needs margins. L■ is a direct frequency readers can audit by hand on small samples.

11 / REPO USAGE

Daily majors workflow

1. Freeze margins before tail counts

Use one ranked or filtered pipeline per experiment version in experiments/v001/ when built.

2. Pre-specify α and tail direction

Lower, upper, or pairwise opposite tails — document before scanning pairs.

3. Store counts, not only ratios

Export numerator and denominator CSV columns alongside L■ so reruns are comparable.

Load snapshots from data/snapshots/; do not open MT5 from this folder.

Software outline

```
joint = 12
cond = 20
L_hat = joint / cond # 0.60
```

In Python, build u,v arrays, then boolean masks (u <= alpha) & (v <= alpha) for the numerator and (v <= alpha) for the denominator. Copula packages estimate parametric λ; this guide emphasizes transparent counting first.

Align dates across majors; a tail in one pair on a holiday mismatch breaks pairs silently if you merge carelessly.

Limits

Small denominators make L■ unstable. Changing α or margin models changes counts. Regime shifts mix crises with calm periods unless you slice time.

Tail dependence is not Granger causality and not a hedge ratio. Joint tail days may share a common shock (USD factor) rather than one pair “causing” the other.

L■ at α = 0.05 is a practical FX research cutoff; it is not the same object as an extrapolated λL unless you model the limit explicitly.

Practice

Q: Is L■ = 0.60 the same as “60% of all days”? **A:** No — it is 60% of conditioning-tail dates (here 12/20).

Q: Can u,v come from price levels? **A:** Use return margins, not trending prices.

Check: 12/20 = 0.60 at $\alpha = 0.05$ in the teaching table.

Q: Does a high Pearson r prove high L■? **A:** No — tails can differ from the center of the distribution.

Sources

[1] Patton, A. J. (2006). Modelling asymmetric exchange rate dependence. International Economic Review, 47(2), 527–556.
FX asymmetric tail motivation.
Open source

[2] forex-formulas-explained.pdf — section 19 (this repository).
Project L■ and λL definitions.
Open source

[3] 18 - Copulas — related u,v margin workflow.
Copula tail parameter context.
Open source

Rebuild: python build_guide.py. End of tail dependence beginner guide.